WORKED SOLUTION

STEP II 1994, Question 9

Mechanics • Equilibrium of Rigid Bodies

STEP II 1994, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q9

The key idea

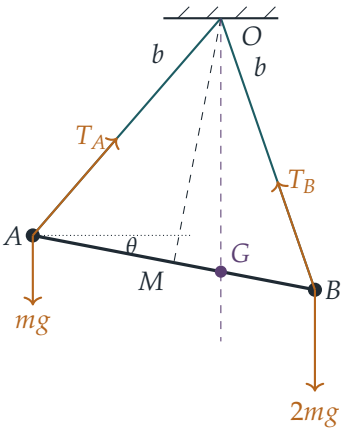
Treat the rod, the two strings and the two particles as one rigid body. The only external forces on it are the two weights and the support at O, so for equilibrium the combined weight, acting at the centre of mass G, must act along a line through O: G hangs vertically below O. That turns the angle into pure geometry of an isosceles triangle. For the tension, take moments about B, which removes both unknown forces acting at B.

Solution

Model. The rod and strings are light, the strings inextensible, and A, B are particles. Since $OA = OB = b$ and $AB = 2a$, triangle OAB is isosceles. Let M be the midpoint of AB; then $OM \perp AB$ and, by Pythagoras,

$$OM = h = \sqrt{b^2 - a^2}.$$

The heavier particle B hangs lower, and the rod makes angle θ with the horizontal.



The angle

Consider the whole system: rod, strings and particles together. The external forces on it are the weights mg at A and $2mg$ at B, and the force from the support at O. Taking moments about O, the support force has no moment, so the total moment of the weights about O must vanish. The two weights are parallel, so they combine into a single force $3mg$ through the centre of mass G of the particles. Its moment about O is zero only if G lies on the vertical through O. So G is directly below O.

Now find G. It lies on AB with $AG : GB = 2m : m = 2 : 1$, so $AG = \frac{2}{3} \cdot 2a = \frac{4a}{3}$, and therefore

$$MG = AG - AM = \tfrac{4a}{3} - a = \tfrac{a}{3},$$

with G on the B side of M.

Ask yourself Why is the angle MOG the same as θ?

OG is vertical and OM is perpendicular to the rod. Rotating the vertical onto OM is the same rotation that takes the horizontal onto the rod, so the angle between OG and OM equals the angle between the rod and the horizontal, namely θ.

Triangle OMG has a right angle at M, so

$$\tan\theta = \frac{MG}{OM} = \frac{a/3}{\sqrt{b^2 - a^2}} = \frac{a}{3\sqrt{b^2 - a^2}},$$

as required.

The tension in AO

Again take the rod and particles together. The forces on this system are T_A along AO, T_B along BO, and the two weights. (The force in the rod is internal.) Taking moments about B kills T_B and the weight $2mg$.

The perpendicular distance from B to the line OA is twice the area of triangle OAB divided by OA. That area is $\frac{1}{2} \cdot 2a \cdot h = ah$, so the distance is $2ah/b$. The weight mg at A has moment arm equal to the horizontal distance between A and B, which is $2a\cos\theta$. The two moments have opposite senses, and balancing them gives

$$T_A \cdot \frac{2ah}{b} = mg \cdot 2a\cos\theta \quad \Longrightarrow \quad T_A = \frac{mgb\cos\theta}{h}.$$

From $\tan\theta = \dfrac{a}{3h}$, with θ acute, we get $\cos\theta = \dfrac{3h}{\sqrt{9h^2 + a^2}}$, and $9h^2 + a^2 = 9b^2 - 8a^2$. Hence

$$T_A = \frac{mgb}{h} \cdot \frac{3h}{\sqrt{9b^2 - 8a^2}} = \frac{3mgb}{\sqrt{9b^2 - 8a^2}}.$$

(The same argument with moments about A gives $T_B = 2T_A$: each string tension is proportional to the mass it supports.)

Checks

- If $a \to 0$ the strings become vertical and $T_A \to 3mgb/(3b) = mg$, which is exactly what particle A needs.
- Vertical resolution for the whole system should give $3mg$. With $a = 1$, $b = 2$, $m = 1$, $g = 9.8$ we get $T_A \approx 11.11\,\text{N}$ and $T_B \approx 22.22\,\text{N}$. Resolving these along the string directions gives a total vertical component of $29.4\,\text{N} = 3mg$.
- A numerical check places the rod at the predicted angle for 2000 random (a, b, m) and solves the force balance at each particle separately. The two particles agree on the rod force, which is a thrust, and the tension matches the formula to about 10^{-15}.

Common slips

- Putting G a distance $\frac{2a}{3}$ from M, or on the wrong side. The centre of mass is nearer the heavier particle, at $\frac{a}{3}$ from the midpoint.
- Taking the moment arm of T_A about B to be $2a$ or b. You need the perpendicular distance from B to the line OA.
- Leaving the answer in terms of θ, when it is asked for in terms of m, g, a and b.

Beyond STEP

The principle that a body hanging freely from one point settles with its centre of mass below that point is the simplest case of a deeper fact: stable equilibria minimise potential energy, and here the potential energy is just $3mg$ times the height of G. At university this becomes the virtual work principle and Lagrangian mechanics, where equilibria are found by differentiating an energy rather than by resolving forces. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and energy methods carefully.

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